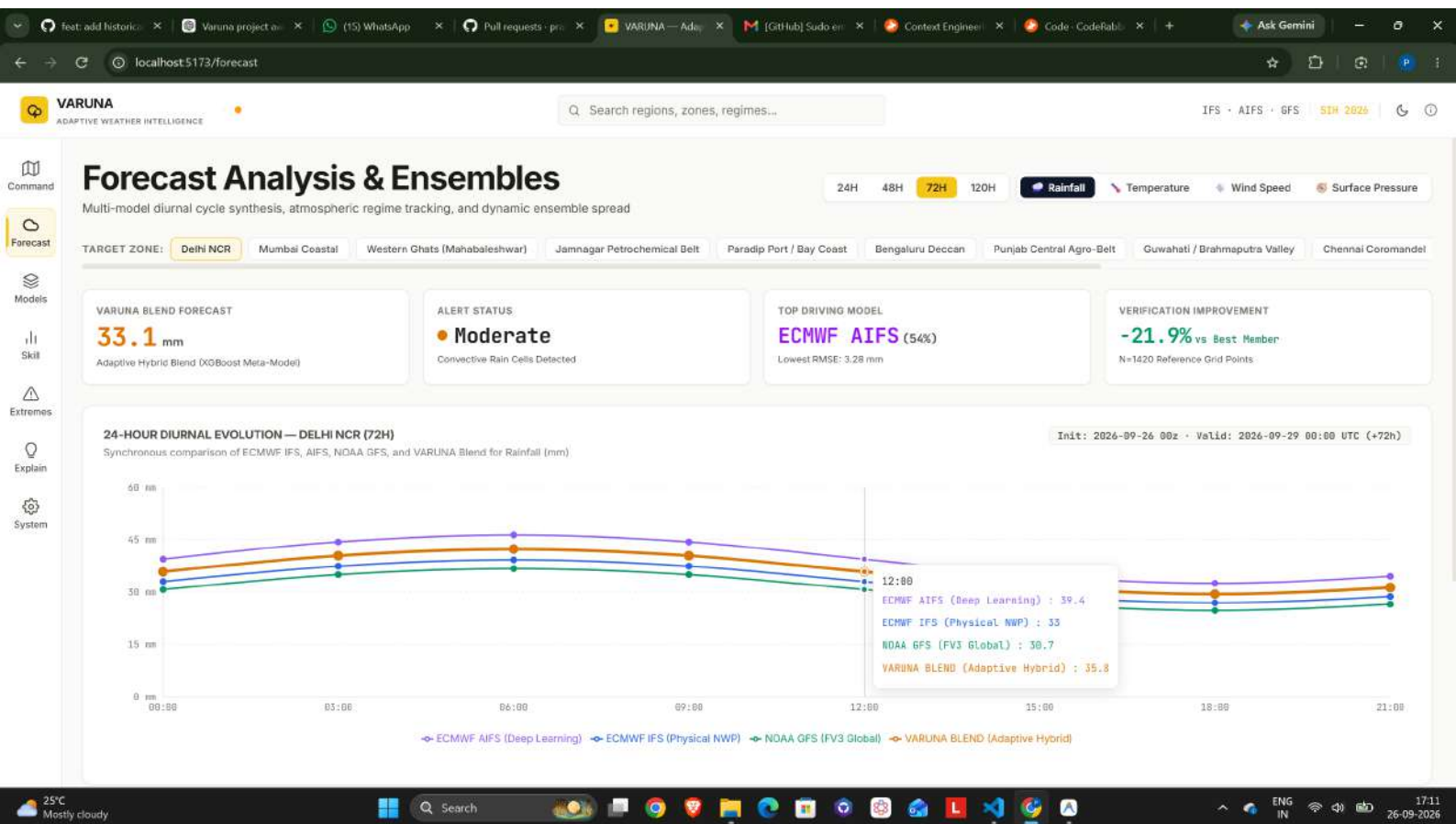




25°C Mostly cloudy

Search

17:11 26-09-2026

VARUNA
ADAPTIVE WEATHER INTELLIGENCE

Search regions, zones, regimes...

IFS · AIFS · GFS · 5TH 2026

Synchronous comparison of ECMWF IFS, AIFS, NOAA GFS, and VARUNA Blend for Rainfall (mm)

Time	ECMWF AIFS (Deep Learning)	ECMWF IFS (Physical NWP)	NOAA GFS (FV3 Global)	VARUNA BLEND (Adaptive Hybrid)
00:00	32	31	30	38
03:00	33	32	31	42
06:00	34	33	32	45
09:00	33	32	31	42
12:00	32	31	30	38
15:00	31	30	29	35
18:00	30	29	28	45
21:00	31	30	29	38

REGIONAL ATMOSPHERIC REGIME PROFILE

Climatological forcing parameters and local topography

North-West Plains

Active Weather Regime

Northern Plains Convective / Western Disturbance

COORDINATES

28.6139°N, 77.2098°E

TERRAIN ELEVATION

216m MSL

IMD AWS — Integration Pending

No verified station observations connected (28 planned stations)

MULTI-MODEL CONSENSUS & WEIGHT CONTRIBUTION

Relative adaptive weights allocated via XGBoost reliability meta-model

XGBoost Kernel

ECMWF AIFS — Deep Learning Transformer

36.3 mm (54%)

NOAA GFS — Global NWP (FV3)

28.6 mm (29%)

ECMWF IFS — High-Resolution Physical NWP

30.6 mm (17%)

VARUNA Weighted Consensus

33.1 mm

25°C Mostly cloudy

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localhost:5173/models

VARUNA
ADAPTIVE WEATHER INTELLIGENCE

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OPERATIONAL MODEL BENCHMARK MATRIX
Target: Delhi NCR • Variable: Rainfall (mm) • Lead Time: 72h
N = 1,428 Reference Verifications

MODEL & ARCHITECTURE	RESOLUTION / GRID	FORECAST (MM)	RMSE (MM)	MAE (MM)	BIAS (MM)	CORRELATION (R)	DYNAMIC WEIGHT	INFERENCE LATENCY
ECHWF IFS Physics-Based NWP	0.1° (~9 km)	30.6	5.72	4.46	+0.58	0.88	17%	42ms
ECHWF AIFS Deep Learning Transformer	0.25° (~28 km)	36.3	3.28	2.49	-0.14	0.95	54%	14ms
NOAA GFS Global NWP Model	0.25° (~28 km)	28.6	4.48	3.58	+0.81	0.85	29%	36ms
VARUNA BLEND Adaptive Hybrid AI-NWP Engine	0.1° Downscaled	33.1	2.56	1.92	+0.04	0.97	100% (Consensus)	8ms

MODEL SKILL SCORE BY WEATHER REGIME (0-100 BENCHMARK)
Verification against reference reanalysis datasets across Indian synoptic regimes (IMD AWS integration pending)
Regime Benchmark

Orographic Precip
ECHWF AIFS (Deep Learning) : 91
ECHWF IFS (9km NWP) : 72
NOAA GFS (FV3 NWP) : 68
VARUNA BLEND (Adaptive Hybrid) : 96

Weather Regime	ECHWF AIFS (Deep Learning)	ECHWF IFS (9km NWP)	NOAA GFS (FV3 NWP)	VARUNA BLEND (Adaptive Hybrid)
Orographic Precip	91	72	68	96
Convective Rain Cells	~85	~65	~75	~90
Thermal Heat Depression	~88	~70	~72	~92
Coastal Gale Influx	~82	~68	~70	~90
Synoptic Pressure Wave	~85	~75	~78	~92

